An HF-ordering over a partial subsystem exists exactly when that subsystem can be made an initial segment

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Editorial note. The counted statement is the author's labelled lemma environment at source0.tex lines 333-335; its proof environment opens at line 336, immediately after \end{lemma}, so the pairing is by adjacency and the retained proof range 336-339 is the whole environment.

Editorial note. This source is newly approved for this line; the preamble is built from the author's own package lines, macro block and theorem-environment declarations (source lines 4-65), with the author's optioned amsmath line omitted so that it cannot clash with the wrapper's amsmath.

Editorial note. The source carries no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label order-over.

Editorial note. Metadata provenance: the source record of the pinned copy and the licence evidence file are both present; the licence is CC BY 4.0 with status supported.

Editorial note. r002 note: r001 failed because the retained author block carried the layout setting mathindent, which only exists when amsmath is loaded with the fleqn option; r002 starts that span at line 22 instead and changes nothing else.

Editorial note. r003 note: the pre-publication gate compares the card's source_version with the source_version recorded in the licence evidence of the same source. This source's evidence file records exactly "2411.13723v3", while the card had carried "arXiv:2411.13723v3 (CC BY 4.0)" copied from the source record; r003 aligns the card with the evidence and changes nothing else. The evidence file itself is not modified.

Editorial scope. One original result of the pinned author manuscript, counted once: the equivalence between the existence of an HF-ordering over a partial subsystem and the possibility of making that subsystem an initial segment (author label order-over).

Allowed standard prerequisites (editorial).

- Partial Steiner triple system. A partial Steiner triple system is a structure with a partial binary operation whose defined products are the blocks, with the Steiner condition on the operation.. It is the structure that carries the orderings.
- HF-ordering. An HF-ordering of B over A is a linear ordering in which the closure of A is A itself and every element is either in A or produces a block whose other members are smaller.. It is the notion the lemma compares with having A as an initial segment.
- Initial segment. A subset is an initial segment of an ordering when every element below an element of the subset belongs to the subset.. It is the second description of the ordering.

Lemma 0.1 Let $A \subseteq B$ be partial STSs. There is an HF-ordering of B over A if and only if there is an HF-ordering of B where A is an initial segment. In fact, given any HF-ordering $<$ of B over A, the sum $(A, <\restriction A) + (B \setminus A, <\restriction (B \setminus A))$ is an HF-ordering of B with initial segment A.

Proof. On the one hand, if $<$ is an HF-ordering of B with initial segment A, it easy to see that it is an HF-ordering over A. On the other hand, assume $<$ is an HF-ordering of B over A. Then the only way in which the sum can fail to be an HF-ordering is for some $a \in B \setminus A$ to be a product of two different pairs of smaller elements, and each element is either smaller than a in $<$ or belongs to A, contradicting that $<$ is an HF-ordering over A. □